QPLog – QC/ON QSO Party Loogbook

Par/By: Bertrand Zauhar, VE2ZAZ - VA2IW
https://ve2zaz.net
https://github.com/VE2ZAZ
https://www.qrz.com/db/VE2ZAZ

Help Manual

Applicable to Version 0.50 or higher, April 2026

Document Index

1 Overview....3
2 Installation and Launch of QPLog....5
2.1 Installation....5
2.2 Launch....5
3 Using QPLog....6
4 Detailed Description....7
4.1 QSO List Window....7
QSO List....8
Buttons Bar....9
“Load Log” Button....9
“New Log” Button....9
"Config Log" Button and Configuration Window....9
“Copy Log” Button....11
"Cabrillo File and Cabrillo File Editing Window" Button....11
"Re-use QSO" Button....12
"Edit QSO" Button....12
"Delete QSO" Button....13
"Delete All QSOs" Button....13
"History" Button....13
"Zones" Button....13
"Statistics" Button....13
"English/Français" Button....13
"Help" Button....13
"About" Button....13
4.2 QSO Entry Window....13
4.3 "Statistics et Scores" Window....14
4.4 Contacted Zones and Sponsor Stations Window....15
4.5 Call History Window....16
5 CW Generation....18
5.1 Description of the Controls of the CW_Support Program....18

"Keyer Mode" Buttons....18
"Connect to Serial Port" Button....19
"Edit Macros" Button....19
"CW Operation Mode" Buttons....19
"CW Speed" Selector....20
"CQ Repeat" Buttons....20
CQ Repeat "Delay" Selector....20
5.2 Details on Operating Modes....20
"WinKeyer" Mode....20
"DTR Line" Mode....20
"RTS Line" Mode....21
5.3 Definition of CW Macros....21
5.4 Details of CW Operation in QPLog....22
Enter (Return) Key....22
Run Mode (CQ Calls)....23
Search and Pounce Mode (Responding to CQ)....23
Escape Key....23
Function Keys F1 to F12....23
5.5 Language Selection for the CW_Support Window....23
6 Various Operating Notes....24
7 Legal License....25
8 References....26

1 Overview

QPLog is a PC software that allows the entry and analysis of radio contacts (QSOs) during the following amateur radio "contest": the Québec QSO Party (QCQP) and the Ontario QSO Party (ONQP). The idea behind the creation of this software was to produce an easy-to-install contest environment, simple to operate, and customized for these two contests. This software project was detailed by Georges-André Chaudron (VE2VAB) and Bertrand Zauhar (VE2ZAZ - VA2IW), with the initial version developed by the latter in 2024.

QPLog is designed for simultaneous participation, if desired, in the QCQP and ONQP. Furthermore, it compiles scores on the fly and independently for each of the two parties, while considering eligible contacts in both contests.

QPLog also allows for mobile operation, accounting for changes in the operator's administrative region (zone) during the contest.

Control of a WinKeyer3 CW or the DTR/RTS line of an external serial port is offered. This allows for automation, and therefore simplifies CW operation during the Party.

The Call History Lookup saves you from having to enter the zone for a QSO with stations known to have previously participated in the QSO Party. As such, it also provides a certain level of confirmation regarding the information entered during the QSO.

Once the contest is over, QPLog can produce a Cabrillo format file, which the participant submits to the contest organization to formalize their participation and provide the list of QSOs.

QPLog is available in pre-compiled versions for Windows 32/64 bits (7, 8, 10, 11) and Ubuntu Linux (Ubuntu 22.xx, 24.xx). The Linux version may also work on other distributions; it has been verified to run, for example, on Arch Linux. However, certain drivers may be required.

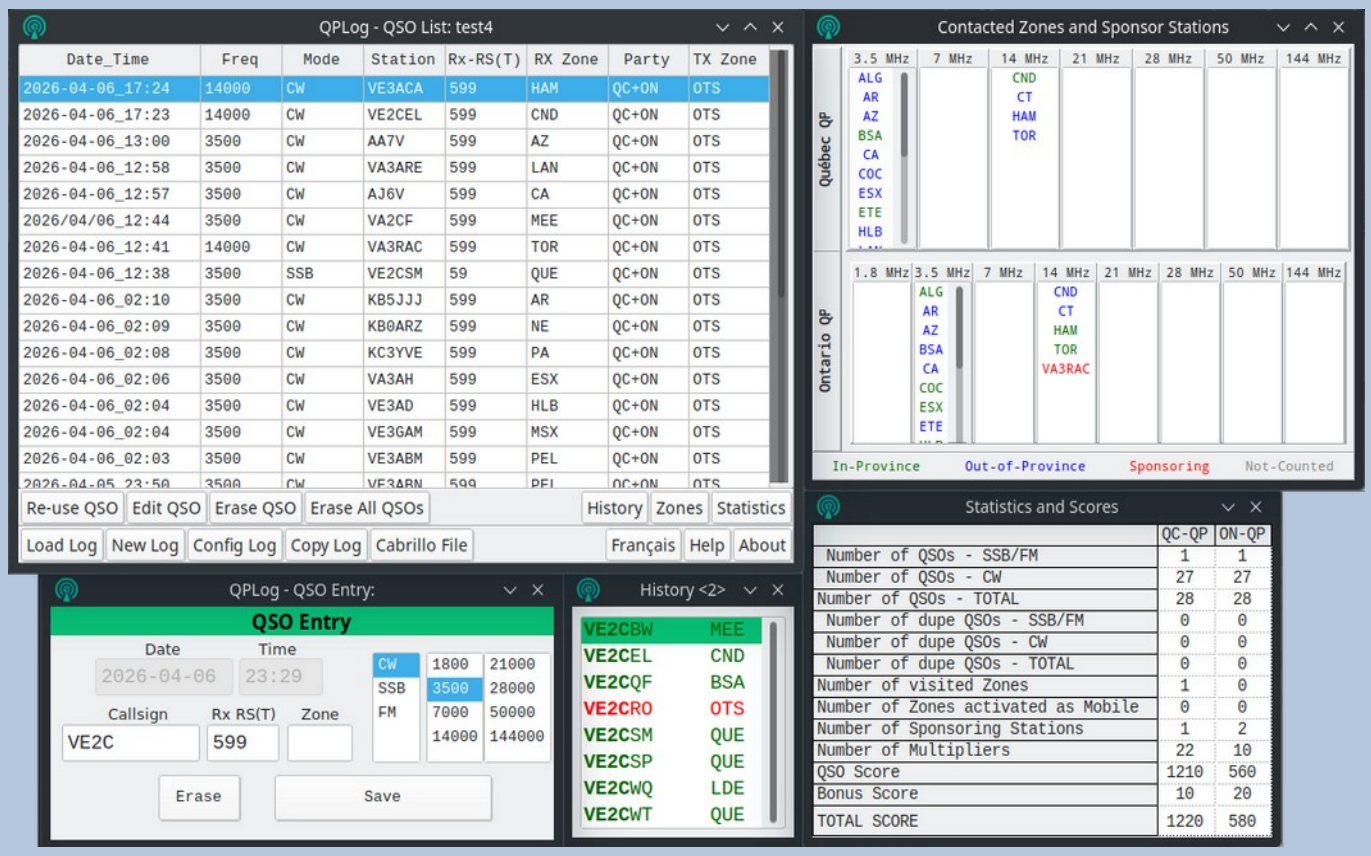


Figure 1: The operations windows of the QPLog software in Kubuntu 24.04

2 Installation and Launch of QPLog

The QPLog software consists of a simple executable file. No changes are made to the operating system directories or the Windows registry. It is also not necessary to obtain administrator privileges to use QPLog.

2.1 Installation

Installation simply involves extracting the contents of the ZIP file into a directory of your choice.
In addition to the program executable files, you will find example Log files in the "Logs" folder. This PDF help file can be found in the "Help" folder.

2.2 Launch

The program file to run is "QSO_Party_Logger" for Linux, and "QSO_Party_Logger.exe" under Windows. A simple double-click on the executable file will launch the program.
If, under Linux, launching the program does not produce a window or only opens briefly, the program may detect an error preventing it from launching correctly. Often, calling the same executable file from a command window will provide additional information about the cause of the failure. Common reasons include outdated or missing required libraries, or, more simply, that the executable file is not marked as executable. In this case, simply enabling the executable flag for the file may resolve the issue. An example of the file properties window and its executable flag is shown below.

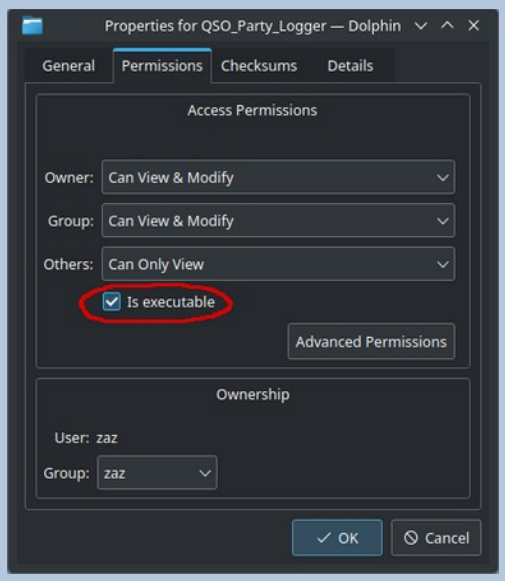


Figure 2: The file properties window in Kubuntu 24.04

3 Using QPLog

The steps for using the QPLog software are intuitive and simple. They follow the typical sequence of a QSO Party.

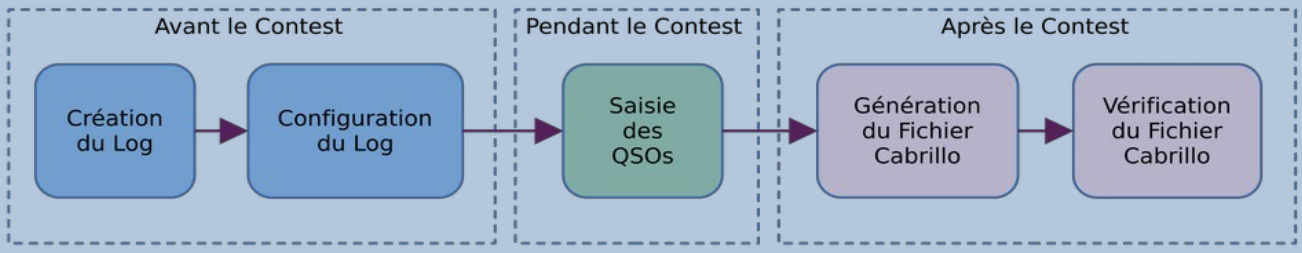


Figure 3: Steps for using QPLog for a QSO Party

In preparation for the QSO Party, the user launches the program. They create a Log using the “New Log” Button . Then they configure various options (choice of QSO Party, participation class, operating zone, etc.) using the "Config Log" Button and Configuration Window. The user can then participate in the contest by entering all QSOs in the QSO Entry Window. Once the contest is finished, the user invokes the "Cabrillo File and Cabrillo File Editing Window" Button, which generates the file and displays it in an editing window. The user can modify its content if needed, and save it. A Cabrillo file is produced for each of the two QSO Parties the user participated in, if applicable. All that remains is to send the Cabrillo file(s) to the organizers of the QSO Parties.

4 Detailed Description

QPLog is displayed across five windows. Two of these are mandatory and always visible: The QSO List and QSO Entry. The other three windows are informational and displayed optionally: The Contacted Zones and Sponsor Stations, the Call History and the Statistics and Scores.

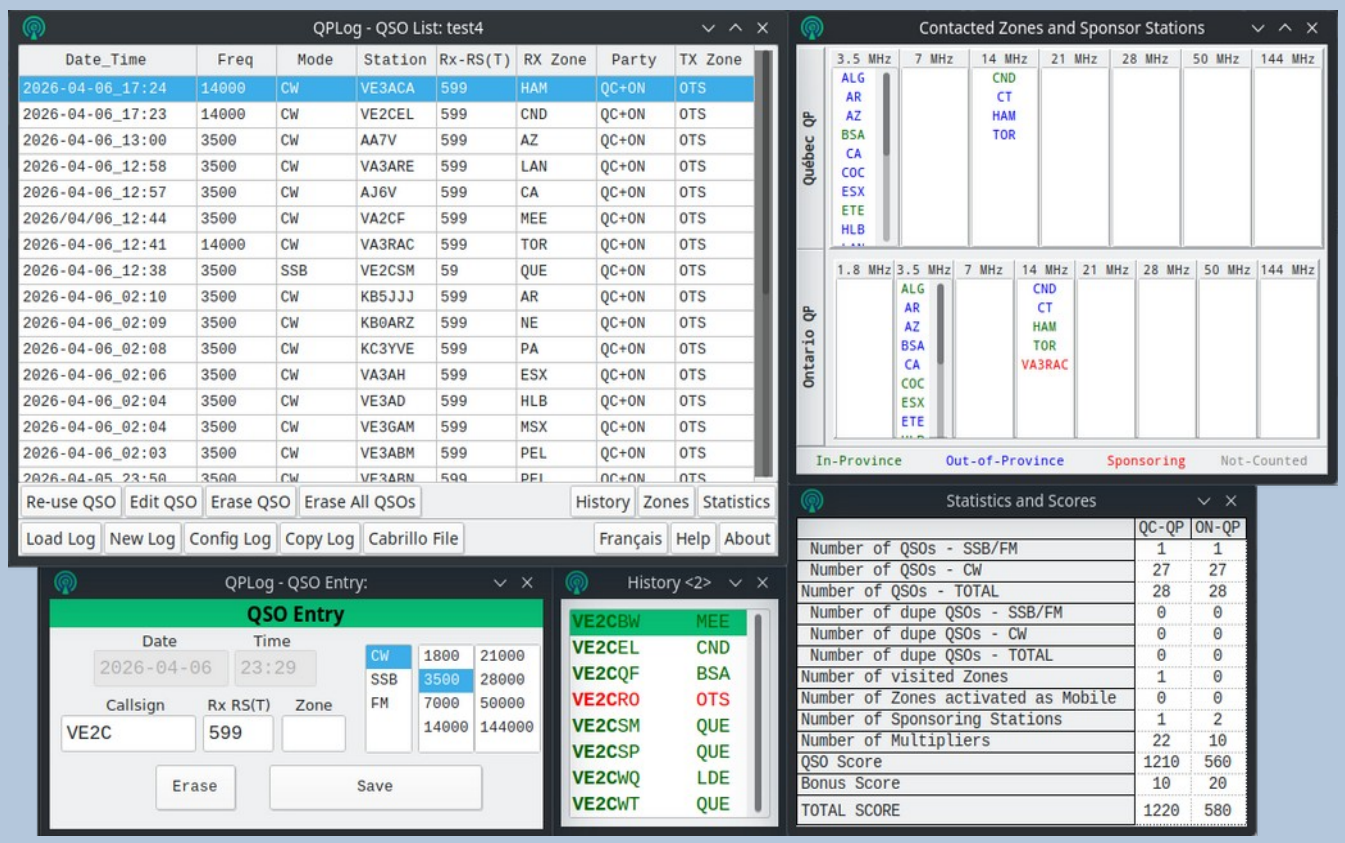
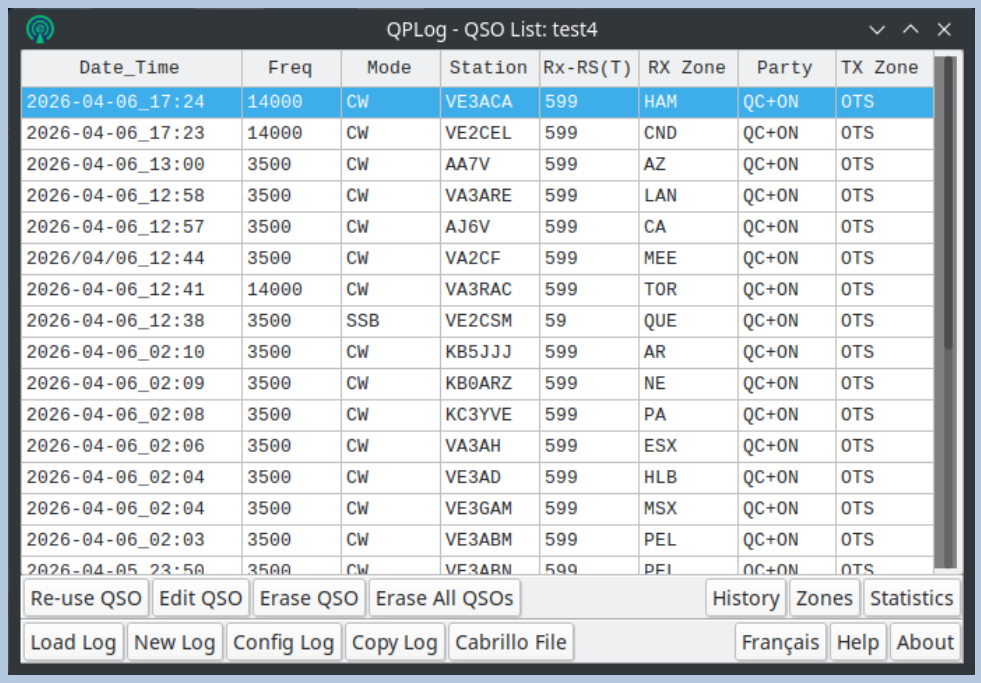


Figure 4: The main windows of the QPLog software

4.1 QSO List Window

The "QSO List" window is the main window of QPLog. In addition to displaying the list of QSOs, it offers a series of buttons for taking action before, during, and after the contest.



Date_Time	Freq	Mode	Station	Rx-RS(T)	RX Zone	Party	TX Zone
2026-04-06_17:24	14000	CW	VE3ACA	599	HAM	QC+ON	OTS
2026-04-06_17:23	14000	CW	VE2CEL	599	CND	QC+ON	OTS
2026-04-06_13:00	3500	CW	AA7V	599	AZ	QC+ON	OTS
2026-04-06_12:58	3500	CW	VA3ARE	599	LAN	QC+ON	OTS
2026-04-06_12:57	3500	CW	AJ6V	599	CA	QC+ON	OTS
2026/04/06_12:44	3500	CW	VA2CF	599	MEE	QC+ON	OTS
2026-04-06_12:41	14000	CW	VA3RAC	599	TOR	QC+ON	OTS
2026-04-06_12:38	3500	SSB	VE2CSM	59	QUE	QC+ON	OTS
2026-04-06_02:10	3500	CW	KB5JJJ	599	AR	QC+ON	OTS
2026-04-06_02:09	3500	CW	KB0ARZ	599	NE	QC+ON	OTS
2026-04-06_02:08	3500	CW	KC3YVE	599	PA	QC+ON	OTS
2026-04-06_02:06	3500	CW	VA3AH	599	ESX	QC+ON	OTS
2026-04-06_02:04	3500	CW	VE3AD	599	HLB	QC+ON	OTS
2026-04-06_02:04	3500	CW	VE3GAM	599	MSX	QC+ON	OTS
2026-04-06_02:03	3500	CW	VE3ABM	599	PEL	QC+ON	OTS
2026-04-05_23:50	3500	CW	VE3ABN	599	PEL	QC+ON	OTS

Figure 5: The “QSO List” window

QSO List

All relevant information for each completed QSO is shown in this window, one line per QSO. Besides the date and time, frequency, mode, contacted station callsign, received signal report, and the contacted station's zone, the table displays the following information:

- The "Party" column displays the party(ies) the user is participating in, as selected in the configuration window.
- The "Zone Tx" column displays the operating zone (administrative region, county) of the user, as selected in the configuration window. If the user moves from one zone to another during the Party, the operating zones are displayed in this column and associated with the QSOs completed in each zone. The user changes the zone during the Party via the configuration window.

The QSO list offers the following functions:

- Sorting alphabetically by column is enabled by clicking on the desired column header. Repeated clicks on a particular column header will toggle between normal and reverse order. A small green triangle will indicate the sorting column and order.
- The user can right-click on a specific QSO to display a dropdown menu offering a series of actions applicable to the QSO (re-use, edit, or delete the QSO).
- Keyboard shortcuts are implemented. The 'CTRL-E' key combination allows editing the selected QSO in the list to the entry window. The 'CTRL-R' combination retrieves the selected QSO's information to the entry window for re-use in a new QSO with the same station. More details on editing and re-using actions are available in the Buttons Bar section below.

Buttons Bar

Two rows of buttons are offered at the bottom of the "QSO List" Window. The top row offers actions taken during the contest affecting the QSOs, while the bottom row groups actions affecting the contest (preparation and completion).

“Load Log” Button

This button opens a window for selecting an existing Log to load into the program. Only QPLog type logs (".qplog" extension) can be opened.

“New Log” Button

This button opens a window to create a new log to load into the program. Once the file name and location are provided, an information window will prompt the user to configure the new log. The log configuration window will open as well. More details on log configuration are provided in the "Config Log" Button and Configuration Window section below.

"Config Log" Button and Configuration Window

This button opens the configuration window for the currently loaded log. Several fields must be filled out/configured before proceeding with QSO entry. Once a first QSO is entered, some fields affecting the contest progress cannot be changed.

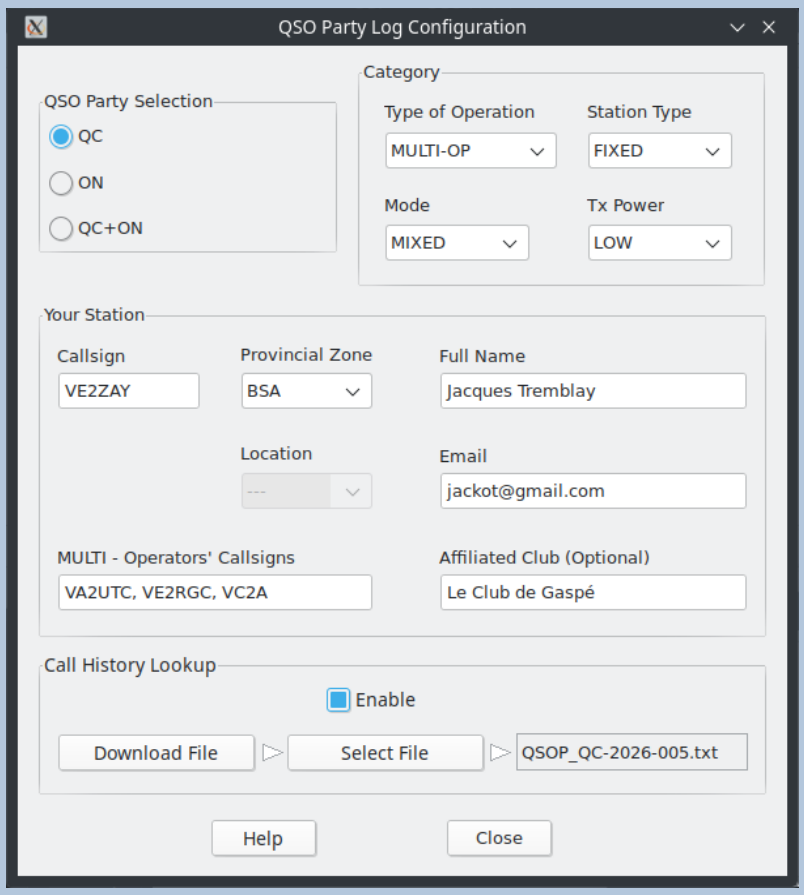


Figure 6: The “Log Configuration” window

The fields offered in the configuration window are:

- **QSO Party Selection:** The user must choose which contest(s) they want to participate in. The choices are:
  - QC: Participation in the Québec QSO Party only. QSOs with Ontario stations can be entered with the "ON" zone or the three-letter Ontario counties.
  - ON: Participation in the Ontario QSO Party only. QSOs with Québec stations can be entered with the "QC" zone or the three-letter Québec administrative regions.
  - QC+ON: Simultaneous participation in both the Québec QSO Party and Ontario QSO Party. QSOs with Québec and Ontario stations must be entered with their administrative region or three-letter county. Any QSO entered with the "QC" or "ON" zone will be excluded from the associated Party score.
- **Category**:
  - Operation Type: SINGLE-OP (one participant), MULTI-OP (multiple participants forming a team under the same callsign), CHECKLOG (the participant will provide their log to the contest organizers for analysis but does not wish to appear in the final ranking).
  - Station Type: FIXED (station stays at the same location throughout the contest), MOBILE (station moves, possibly changing zones, during the contest).
  - Mode: CW (only CW mode during the contest), SSB / FM (phone mode only during the contest), MIXED (using both CW and phone during the contest).
  - TX Power: HIGH (can use full legal power), LOW (limited to 100 Watts output), QRP (limited to 5 Watts output).
- **Your Station**:
  - *Callsign:* Your callsign, or the callsign used by a group.
  - *Provincial Zone:* For Québec stations, select your administrative region from the 17 official regions. For Ontario stations, select your county from the 50 recognized counties. For stations outside Québec and Ontario, leave the provincial zone selection as “----,” which activates the "Location" dropdown menu. For "mobile" operation, the zone can be changed during the Party as you move around.
  - *Location:* For stations outside Québec and Ontario, select your recognized ARRL section. For Québec or Ontario stations, leave the location selection as “----,” which activates the "Provincial Zone" dropdown menu.
  - *Full Name, Email, and Affiliated Club:* Required fields for producing the Cabrillo file.
- **Call History Lookup**
  - *Activation* Checkbox: This checkbox enables the Station History Lookup function.
  - *Download File* Button: This button directs the computer's default web browser to the Station History file download page.

- *Select File* Button: This button allows you to select the Station History file downloaded locally to the computer. The name of the selected file is displayed in the box to the right of this button as confirmation.

For more details on how the Station History function works, see the Call History Window section below.

“Copy Log” Button

This button allows making a copy of the currently loaded log under a different name. Note that the original log remains intact, and the copied version is now loaded into memory.

"Cabrillo File and Cabrillo File Editing Window" Button

This button generates a Cabrillo file for each of the two QSO Parties, if applicable. The location of the generated Cabrillo files is the same as the location of the log file (.qplog).

As the first step, a "SoapBox" comment entry window appears on the screen for each of the two activated QSO Parties, one after the other. If desired, the user can write comments, such as their opinion of the contest or their evaluation of propagation during the contest. Whether or not comments are added, the user clicks "Continue" to proceed to the next step.

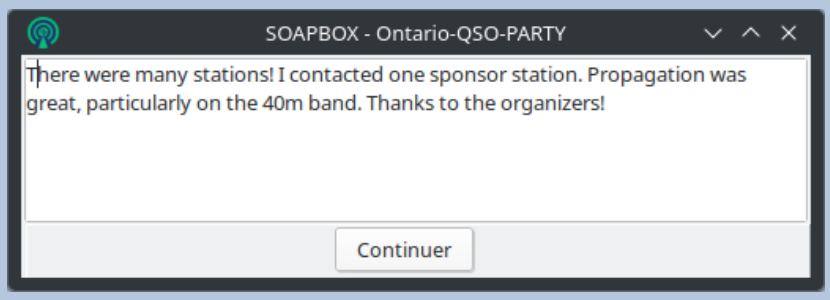


Figure 7: The "SoapBox" comment window

Once the comment entry is completed, an editing window for the Cabrillo file will appear for each of the two activated QSO Parties. If the Cabrillo content is correct and does not need changes, the display/editing window can simply be closed by clicking the "Close" button. If alterations are necessary, they can be made directly in the editing window. The changes will be saved when the "Save Changes" button is clicked.

Note that the header content is partially drawn from the information provided in the Configuration Window (see "Config Log" Button and Configuration Window for more details.

Reminder: YOU ARE RESPONSIBLE FOR THE CONTENT OF THE CABRILLO FILES. Please review them as necessary to ensure they conform to the expected format and content.

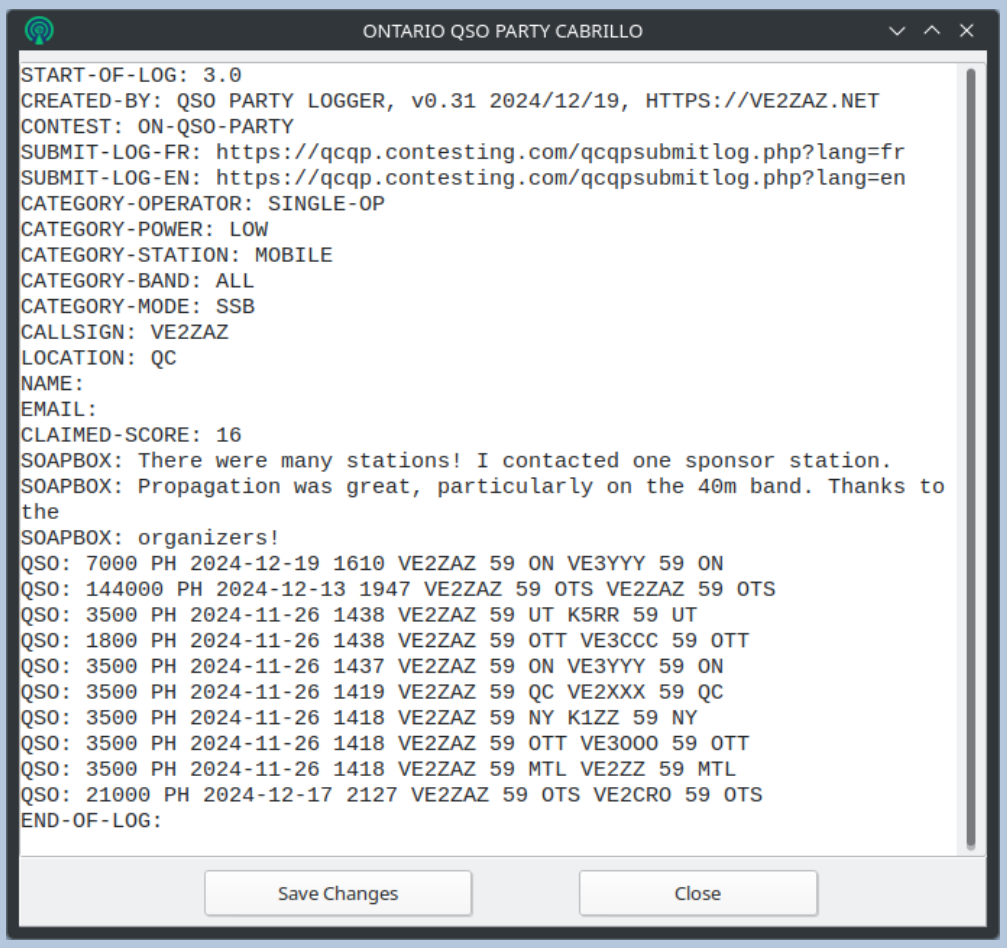


Figure 8: The Cabrillo file edit window

"Re-use QSO" Button

This button allows modifying the information of a previous QSO. The selected QSO's information is transferred to the entry window. The user just needs to modify the QSO information and click the "Edit" button to update the QSO. While editing the QSO, the action banner in the entry window will display “Editing a QSO” in yellow. Duplicate QSO checks are performed during the entry of the QSO modifications.

"Edit QSO" Button

This button allows modifying the information of a previous QSO. The selected QSO's information is transferred to the entry window. The user just needs to modify the QSO information and click the "Edit" button to update the QSO. While editing the QSO, the action banner in the entry window will display “Editing a QSO” in yellow. Duplicate QSO checks are performed during the entry of the QSO modifications. Note that the CTRL-E key combination has the same effect as the Edit QSO button.

"Delete QSO" Button

This button deletes the selected QSO from the QSO list. Only one QSO can be deleted per button action. Note that the action is instantaneous and irreversible.

"Delete All QSOs" Button

This action deletes all QSOs from the QSO list. A confirmation window will appear to prevent accidental deletion. Deleting the QSOs is irreversible.

"History" Button

This button opens the “Call History” window. More details about this window are available in the section: Call History Window

"Zones" Button

This button opens the "Contacted Zones and Sponsor Stations" window. More details on this window are available in the Contacted Zones and Sponsor Stations Window section.

"Statistics" Button

This button opens the "Statistics" window. More details on this window are available in the "Statistics et Scores" Window section.

"English/Français" Button

This button toggles the software's display language between French and English.

"Help" Button

This button opens this help document in PDF format in the default web browser of the computer. The PDF document is also located in the "Help" folder accompanying the software.

"About" Button

This button opens the information window about the software. To close this window, click the "Close" button.

4.2 QSO Entry Window

The QSO entry window allows entering the details of each QSO during the contest. It also allows correcting a QSO already entered in the Log.

The QSO entry window guides the entry of information in multiple ways:

- By default, the date and time entered are the current UTC values provided by the PC. The only way to modify the date or time of a previously entered QSO is to edit it (via the "Edit QSO" Button).
- The available modes (SSB, CW, FM) correspond to the participation class entered in the configuration window.
- The choice of frequencies corresponds to the eligible frequency bands in the selected QSO Parties.

- All characters entered are converted to uppercase.
- Only allowed characters in a callsign can be used.
- For simplicity, the "sent report" field does not appear in the window. The sent report should always be "59" for phone and "599" for CW.
- The spacebar and tab key allow moving from one field to the next in a loop.
- Once all the QSO data is filled in, the "Return" key records the QSO. Note that the function of the "Return" key is modified when external keyer control is enabled. See the Enter (Return) Key section for more details.
- The "Escape" key cancels the entry and clears the fields. Note that the function of the "Escape" key is modified when external keyer control is enabled. See the Escape Key section for more details. Note that the CTRL-W key combination has the same wiping effect, and that it is always available.
- A real-time duplicate (dupe) check is performed with each keystroke. If a duplicate QSO exists, the QSO entry window will turn orange, and the duplicate QSO(s) will appear in orange in the "QSO List" window. The entry window will return to its default color once the entered information no longer matches a duplicate QSO. The duplicate criteria are: same callsign, received zone, mode, and frequency band. Note that it is still possible to enter a duplicate QSO, as long as you acknowledge a warning window beforehand.

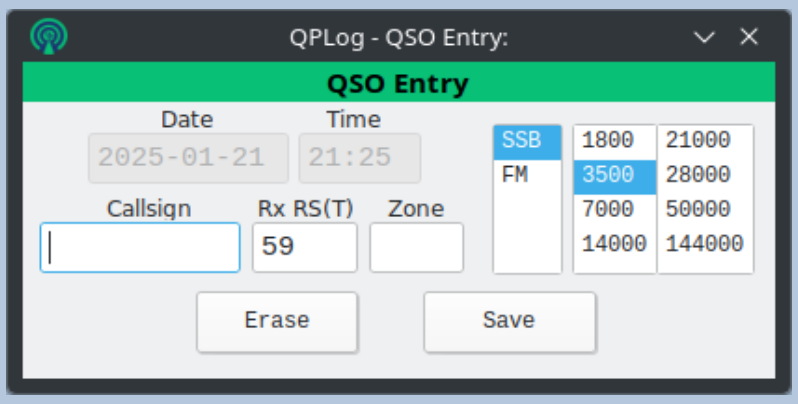


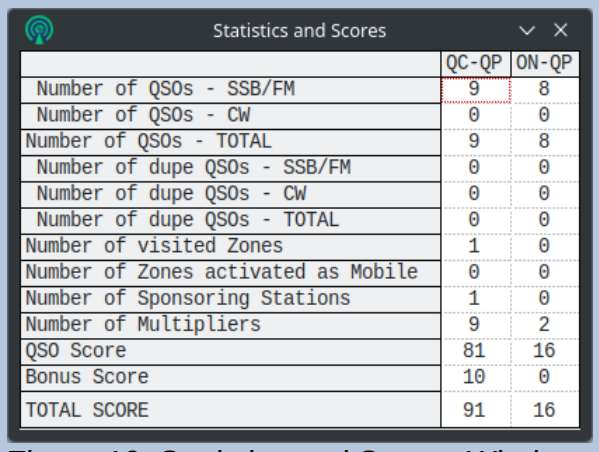
Figure 9: QSO Entry Window

4.3 "Statistics et Scores" Window

The "Statistics and Scores" window displays, in real-time, the statistics and total contest score for each of the two QSO Parties, if applicable. The labels for the rows are quite descriptive.

A distinction is made between the lines "Number of Zones Visited" and "Number of Zones Activated in Mobile," with the latter indicating the operation zones where at least 3 QSOs were completed, thus officially activating the zone.

This window cannot be modified by the user.



Statistics and Scores

	QC-QP	ON-QP
Number of QSOs - SSB/FM	9	8
Number of QSOs - CW	0	0
Number of QSOs - TOTAL	9	8
Number of dupe QSOs - SSB/FM	0	0
Number of dupe QSOs - CW	0	0
Number of dupe QSOs - TOTAL	0	0
Number of visited Zones	1	0
Number of Zones activated as Mobile	0	0
Number of Sponsoring Stations	1	0
Number of Multipliers	9	2
QSO Score	81	16
Bonus Score	10	0
TOTAL SCORE	91	16

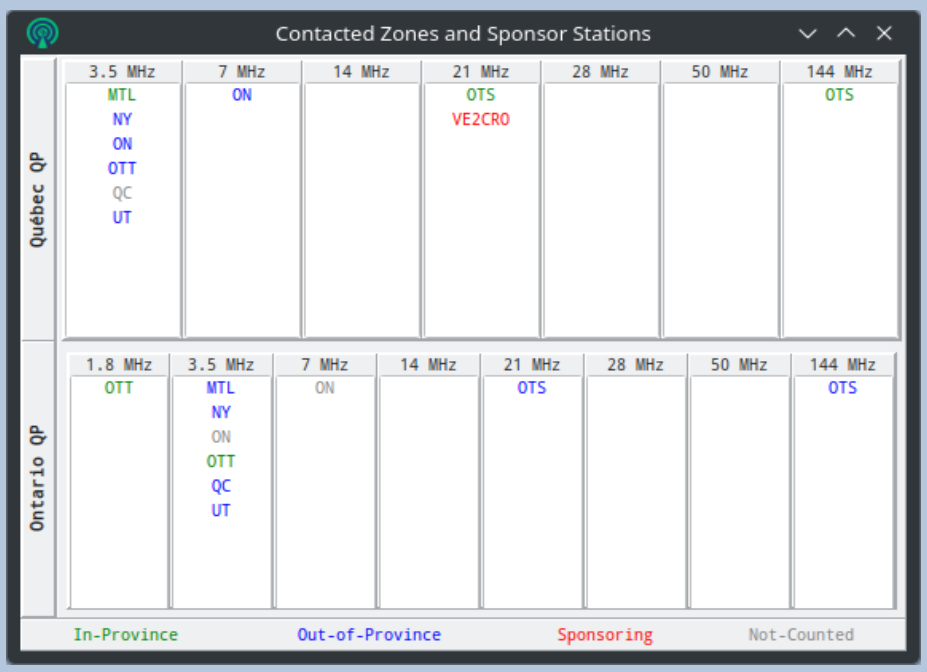
Figure 10: Statistics and Scores Window

4.4 Contacted Zones and Sponsor Stations Window

The "Contacted Zones and Sponsor Stations" window provides, in real-time, the list of contacted zones and locations for each valid frequency band.

A color distinction is made between the zones within the operating province (green) and zones outside the province (purple). Note that non-scored zones are displayed in gray. Additionally, contacted sponsor stations are shown in red. All this information is displayed for each of the two QSO Parties, if applicable.

This window cannot be modified by the user.



Contacted Zones and Sponsor Stations

Québec QP

3.5 MHz	7 MHz	14 MHz	21 MHz	28 MHz	50 MHz	144 MHz
MTL NY ON OTT QC UT	ON		OTS VE2CRO			OTS

Ontario QP

1.8 MHz	3.5 MHz	7 MHz	14 MHz	21 MHz	28 MHz	50 MHz	144 MHz
OTT	MTL NY ON OTT QC UT	ON		OTS			OTS

In-Province Out-of-Province Sponsoring Not-Counted

Figure 11: La fenêtre des Zones et Stations Commanditaires Contactées

4.5 Call History Window

The Call History Lookup feature is convenient but optional. It saves you from manually entering the zone for a QSO with stations known to have previously participated in the QSO Party. As such, it also provides a degree of confirmation regarding the information entered during the QSO.

This window is displayed when the Call History feature is enabled in the Log Configuration window. To display this window, the user clicks the "History" button in the button bar of the "QSO List" window.

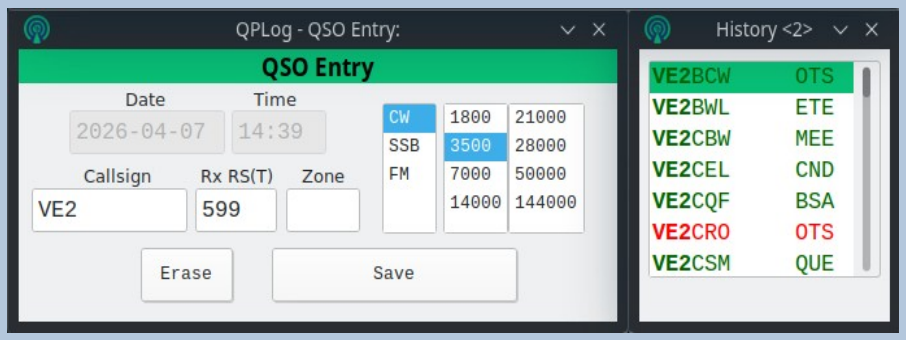


Figure 12: The QSO Entry and Call History windows during a callsign entry

The Call History window displays the results of a search performed in the call history file. The search compares the radio callsign in the "Callsign" field of the QSO Entry window to the contents of the history file. Callsigns with matching characters are displayed in the window. This window is dynamically updated each time the entered call sign characters change. The closer the entered partial callsign is to a valid callsign, the fewer callsigns are shown in the History window. Matching characters are in bold. Furthermore, the colors used in the History window correspond to the colors associated with station locations, as used in the Contacted Zones and Sponsor Stations window.

If a unique callsign is found (only one callsign shown in the list), the additional information available in the History file for that callsign (first name and region) appears for informational purposes below the list in the History window. When the operator is certain they have correctly entered the radio callsign, they press the spacebar, which transfers the zone of the selected station from the History window to the Zone field in the QSO Entry window. The cursor is also automatically repositioned in this field. Although the "Tab" key moves the cursor from one field to the other, it does not transfer the station's zone, unlike the spacebar.

When the operator is certain they have entered the correct radio callsign, they press the spacebar, which transfers the selected station's zone from the History window to the Zone field in the QSO Entry window. The cursor is also automatically repositioned within this field. Although the "Tab" key moves the cursor from one field to another, it does not transfer the station zone, unlike the action of the spacebar.

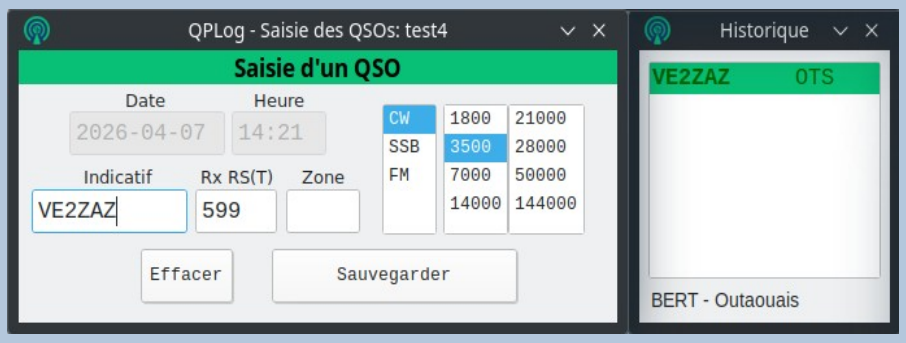


Figure 13: The QSO Entry and Call History windows showing a unique callsign

During callsign entry, the operator can click on one of the stations displayed in the History window, which transfers the selected callsign to the "Callsign" field in the QSO Entry window. The operator can then use the space bar as described above.

The Call History function is active even if the History window is not displayed. This applies provided that the Call History function is enabled in the Log Configuration window and that the selected history file is valid. For more details on enabling the Station History function, see section: "Config Log" Button and Configuration Window.

It is advisable to download and select the most recent Call History file just before holding a QSO Party. This file is available on the N1MM+ software website, the address of which can be found in the References section below. The files for the Quebec QSO Party have a name beginning with "QSOP_QC". The files for the Ontario QSO Party have a name beginning with "QSOP_ON". Note that files list can span over more than one web page.

5 CW Generation

QPLog allows the automation of CW (Morse) code generation, which facilitates operation during the Party and reduces the fatigue associated with manipulating a Morse key or paddle. The program supports an external keyer of the WinKeyer3 type. It can also act as a keyer itself and generate CW code directly by toggling the DTR (Data Terminal Ready) or the RTS (Request To Send) control line of a serial port available on the computer. CW support is handled through an external program that accompanies the QPLog software. The program is named “CW_Support” and is located in the "CW_Support" directory. The CW_Support program must be launched and must run at the same time as QPLog for CW support to function.

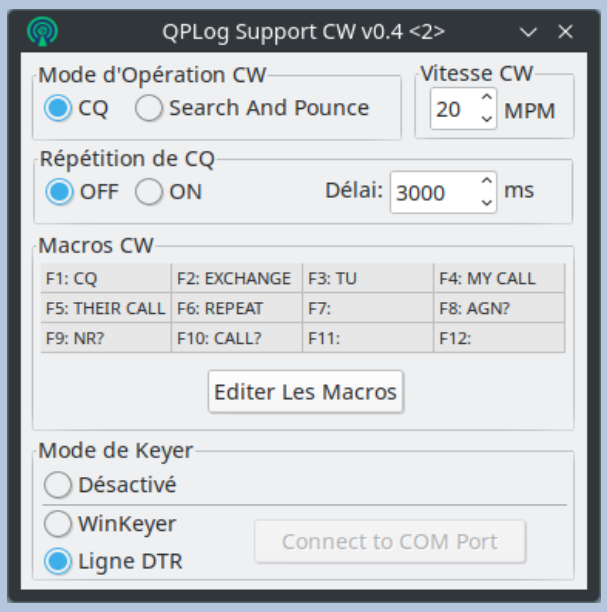


Figure 14: The CW_Support program window

5.1 Description of the Controls of the CW_Support Program

Here is a description of the various buttons and controls of the CW_Support program:

"Keyer Mode" Buttons

Once the program is launched, the first step is selecting the Keyer mode. By default, the Keyer mode is Disabled. The selection of WinKeyer or DTR Line is made by clicking the corresponding button. This enables access to the serial port selection button. Note that to change the keyer mode, the Disabled selection must first be chosen, which “closes” the serial port already in use. The new mode can then be selected. This action makes the Serial Port Connection button accessible again. For more details on the WinKeyer mode and the DTR Line mode, consult the section Details on Operating Modes.

"Connect to Serial Port" Button

The Serial Port Connection button opens a configuration window allowing the selection of the serial port (COMxx on Windows, ttyXX on Linux) corresponding to the selected keyer mode. Although several parameters can be modified in the port configuration window, only the first two selections — the “Port” and the transmission speed “Baud Rate” — should normally be changed. Note that the WinKeyer normally operates at a speed of 1200 bps, which the Baud Rate field proposes by default. Also note that the Baud Rate transmission speed is not relevant in DTR Line mode; only the selection of the serial port matters.

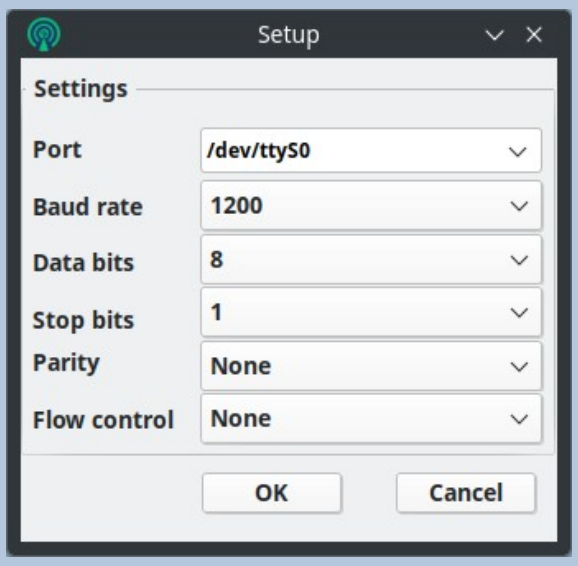


Figure 15: The Serial Port configuration window.

"Edit Macros" Button

CW messages are sent either by pressing the function keys *F1* to *F12* on the keyboard or by pressing the Enter (Return) key. These messages are defined in a text file called “CW_Macros.txt” located in the same directory as the CW_Support program. The labels of these messages are displayed in the function key table in the main window. An “Edit Macros” button located below the table provides easy access to the file so that it can be viewed or modified. The default ASCII text editor of your operating system (for example Notepad in Windows) is launched and the file can be modified as needed. If modifications are made, do not forget to save the file before closing the editor, otherwise the changes will be lost. For more details on the content of the CW macros file, consult the section Definition of CW Macros.

"CW Operation Mode" Buttons

Experienced HF contest operators recognize these two CW operating modes. The Run mode (or CQ calling) is chosen when the operator wishes to send a series of CQ calls, thus initiating QSOs. The Search and Pounce mode (or responding to CQ calls) is selected when the operator wishes to answer CQ calls from other stations. The selected mode determines the content and sequence of automated messages, therefore the CW macros that are sent.

"CW Speed" Selector

This selector sets the CW sending speed in Words Per Minute. The selected speed follows the universally recognized standard. The sending speed can be modified at any time, even while a message is being sent. Note that the sending speed set in the WinKeyer before launching the program cannot be read by the program when it is launched. The operator must therefore set the speed again within the program so that the displayed value matches the value set in the WinKeyer.

"CQ Repeat" Buttons

In Run mode (CQ Calls), CQ calls can be repeated without operator intervention. When the Repeat selection is set to “ON”, the content of function key F1 (the CQ call message) is repeated after a delay determined by the delay selector (see below). This repetition continues until either the Escape key is pressed or characters are typed into the callsign field in the QSO Entry window. Setting Repeat to “OFF” disables CQ repetition.

CQ Repeat "Delay" Selector

The “Delay” selector represents the listening time between two repeated CQ calls. The delay is specified in milliseconds. For example, a value of 3500 represents 3.5 seconds. Note that this selector has no effect when the selected operating mode is Search and Pounce. It also has no effect when the CQ repeat button is set to “OFF”.

5.2 Details on Operating Modes

The program offers two operating modes: “WinKeyer” and “DTR Line” on COM Port. This selection is made when connecting to the serial port.

"WinKeyer" Mode

When WinKeyer mode is selected, the CW_Support program controls an external WinKeyer3 CW keyer (as defined by K1EL) via a USB port that appears as a serial port in the operating system.

Note that the WinKeyer3 implementation allows CW speed changes from an external program. However, it does not allow reading the current CW speed setting of the WinKeyer. When the CW Speed Selector is used to change speed, the CW_Support program therefore assumes that the speed sent to the WinKeyer is adopted by the latter.

For more details on the WinKeyer3, refer to the technical documentation in the References section.

"DTR Line" Mode

When “DTR Line” mode is selected, CW_Support produces CW code by controlling two pins of a serial port (RS-232C) available on the computer. The DTR pin corresponds to the action of a straight Morse key. A high logic level (positive voltage) corresponds to transmitting the RF carrier, while a low logic level (negative voltage) corresponds to no RF carrier. Another serial port pin, RTS, is set high at the beginning of a transmission and returned low at the end of the

transmission. If desired, the PTT line of the transceiver can therefore be controlled by the RTS pin. A short delay is inserted by the program between the switching of the two pins.

The DTR and RTS pins of an RS-232C serial port must not be connected directly to the radio transceiver, as this could damage the transceiver. An interface must be inserted between the computer and the transceiver to convert logic levels into contact closures. Such an interface is quite simple to build. The simplest version contains only a transistor along with a resistor and a protection diode at its base. Consult Internet resources for additional information on this type of interface.

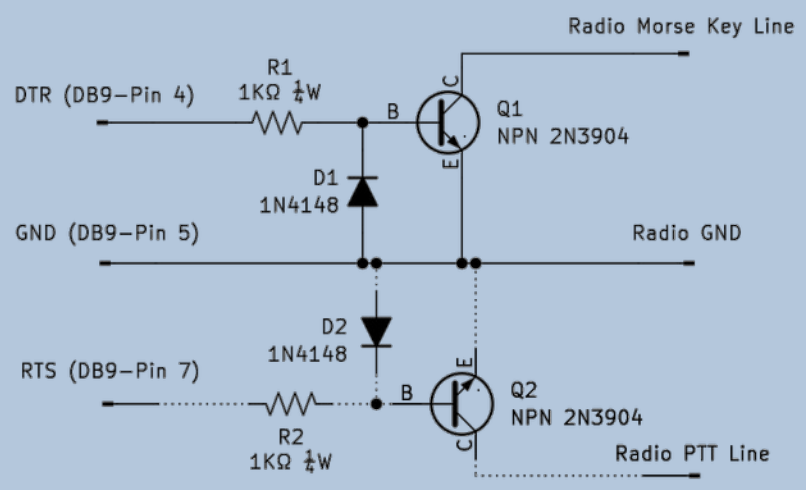


Figure 16: An example of an interface for sending CW via an RS-232C port equipped with a DB9 connector. The lower part shown in dotted lines is assembled when PTT line control is required.

"RTS Line" Mode

When the “RTS Line” mode is selected, the action is the same as with the DTR Line mode above, except that the functions of the two RS-232C pins are inverted: The RTS pin corresponds to the action of a straight Morse key, and the DTR pin controls the transceiver PTT line.

5.3 Definition of CW Macros

The CW messages sent during CW operation are defined as macros in the text file called “CW_Macros.txt”. This file must be located in the same directory as the CW_Support program. It can be viewed and/or modified using an ASCII text editor such as Notepad. In addition to the macros themselves, this file contains detailed explanations of the required syntax. Any deviation from the permitted syntax could lead to incorrect operation of the program or even loss of Party data.

Macros are associated with function keys *F1* to *F12*. Two banks of macros are defined, corresponding to the two operating modes: Run, and Search and Pounce. Here are two macro examples. Note that the fields in a macro must be separated by commas.

```
1F1,CQ,CQ CQ {MYCALL} TEST
```

```
2F4,MY CALL,{MYCALL}
```

Field definitions:

• `mFx` : Identifier of the function key for the two operating modes.
m is ‘1’ for Run mode, or ‘2’ for Search and Pounce mode, followed by ‘Fx’ corresponding to the function key. Never modify these identifiers. Do not insert any character or space before the macro definition.

• `LABEL` : The label of the message displayed in the function key table in the main program window. The label should be short enough to be visible in the table.

• `MESSAGE` : The CW message to transmit.

Three predefined variables may also be used in the CW message. These variables insert information obtained from QPLog at the time of the QSO.

• `{MYCALL}` : Replaced by your callsign, taken from the Setup window of QPLog.
• `{THEIRCALL}` : Replaced by the callsign of the other station currently entered in the callsign field.
• `{EXCH}` : Replaced by your Zone, taken from the Setup window of QPLog.

The variables must include the braces in the message, as shown in the examples above.

Macros 1F1, 1F2, 1F3, 2F1 and 2F2 have pre-assigned uses and their content should only be modified with their role in mind during QSO entry operation. See the section Details of CW Operation in QPLog for more details.

The other function keys may be modified at the operator’s discretion.

5.4 Details of CW Operation in QPLog

Once the CW_Support program is launched and configured, CW operation is performed from within QPLog. The action takes place in the QSO entry window. The keys associated with CW operation are Enter (Return), Escape, and function keys *F1* to *F12*. Their actions are identical whether operating in WinKeyer mode or DTR Line mode.

Enter (Return) Key

The Enter key has multiple actions depending on the stage of the QSO and the operating mode. Combined with the Space bar and Escape key, the Enter key allows continuous CW operation without using the mouse or removing hands from the keyboard.

The Enter key generates specific actions according to the current operating mode.

Run Mode (CQ Calls)

When the cursor is in the callsign field, the Enter key sends the contents of macro 1F1, the CQ call. If CQ repetition is enabled, CQ calls will automatically repeat and will stop when characters are entered in the callsign field or when the Esc key is pressed. If three or more characters are present in the callsign field, the program considers the callsign as received, In such case, the Enter key sends the content of macro 1F2, the report (“599 Zone”). At the same time, the cursor automatically moves to the Zone field in preparation for receiving the other station’s report. When the callsign has been received and the Zone field contains two or more characters, the QSO is considered completed. In such case, the Enter key sends macro 1F3, the thank-you message along with an invitation to the next station (“TU VE2ZAZ TEST”), and then saves the QSO.

Search and Pounce Mode (Responding to CQ)

When the cursor is in the callsign field, the Enter key sends macro 2F1, which transmits the operator’s callsign. When the operator considers the other station’s callsign as having been correctly received, they move the cursor to the Zone field (Space bar, Tab key, or mouse click) to receive the report (Zone). When the Zone field contains at least two characters, pressing Enter sends macro 2F2, the report (“TU 599 Zone”), and then saves the QSO.

Escape Key

The Escape (Esc) key always interrupts the sending of CW macros, regardless of the situation. Pressing Escape does not disable CQ repetition. At the next CQ call, repetition will still be active. Note that Escape reverts to its normal function of clearing input fields when QPLog does not detect the presence of the CW_Support program.

Function Keys F1 to F12

Function keys *F1* to *F12* provide an alternative way to operate CW and allow sending additional requests or exchanges. They send the macros defined for each key in the file “CW_Macros.txt”. In Run mode, macros 1F1 to 1F12 are sent. In Search and Pounce mode, macros 2F1 to 2F12 are sent. It is therefore possible to operate CW entirely using the function keys. Moreover, the CW function keys are functional in all operations windows withing QPLog.

When QPLog does not detect the presence of the CW_Support program, keys *F1* to *F12* have no defined function.

5.5 Language Selection for the CW_Support Window

The display language (French or English) in the CW_Support program is selected when it is launched. The program reads the configuration file of QPLog and applies the same language selection as QPLog. If the QPLog configuration file is not found, French is used as the default language.

To update the display language in CW_Support after having changed the language in QPLog, it is necessary to exit and then restart the program.

6 Various Operating Notes

- The log content is updated and saved after every QSO entry, modification, or deletion. Therefore, no manual saving is required during the contest.
- A backup copy of the log file is created in the same directory as the ".qplog" log file. The backup file is created before any action on the log. This file has the ".qplog_bak" extension. This backup file could serve as a "last resort" to undo an unwanted action on the log, but it requires manual intervention outside of the program to correctly replace/rename the ".qplog_bak" file as the ".qplog" file.
- The log configuration is saved in the log file with the ".qplog_cfg" extension. This file accompanies the ".qplog" file and should not be manually modified by the user.
- A program configuration file is created the first time the program is run and updated every time the program exits. The configuration file is named "Session.cfg" and is located in the same directory as the program file. The user should not modify this file at any time.
- It is important to ensure that the PC on which QPLog is running has its date and time correctly adjusted. The validity of the log depends on this.
- The entry of duplicate QSOs, while undesirable, is not penalizing in terms of contest score. This is why the software allows duplicates.
- Entering a three-letter zone from the other province (instead of "ON" during the QCQP) is supported by QPLog to allow the entry of multiple QSOs from mobile stations traveling between different zones within the other province.
- Mobile operation is supported by QPLog. To allow this, the operating zone can be changed during the Party via the configuration window as the operator moves. The operating zone will be displayed for each completed QSO in the QSO list, under the "Tx Zone" column.

7 Legal License

This program is free software: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 3 of the License, or (at your option) any later version. A mention of the program's author would, however, be appreciated in any project derived from it.

This program is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

See the GNU General Public License for more details. Consult https://www.gnu.org/licenses/.

Ce programme est un logiciel gratuit et libre de droits: vous pouvez le redistribuer et/ou le modifier selon les termes de la Licence Publique Générale GNU telle que publiée par la Free Software Foundation, soit la version 3 de la Licence, soit (à votre choix) toute version ultérieure. Une mention de l’auteur du programme serait cependant appréciée dans tout projet découlant de celui-ci.

Ce programme est distribué dans l'espoir qu'il soit utile, mais SANS AUCUNE GARANTIE ; sans même la garantie implicite de COMMERCIALISATION ou d'ADAPTATION À UN BUT PARTICULIER.

Voir la Licence Publique Générale GNU pour plus de détails. Consultez https://www.gnu.org/licenses/.

8 References

The Cabrillo Specification V3:
https://wwrof.org/cabrillo/

Québec QSO Party:
https://quebecqsoparty.org/

Ontario QSO Party:
http://www.va3cco.com/oqp/

VE2ZAZ Site Web / Web Site:
https://ve2zaz.net

The N1MM+ software website from which the Station History files are retrieved:
https://n1mmwp.hamdocs.com/mmfiles/categories/callhistory/
The files for the Quebec QSO Party have a name beginning with "QSOP_QC".
The files for the Ontario QSO Party have a name beginning with "QSOP_ON".